5.9.

$CPi = 1$

mem reference / instr = 1,35

$CPi_{real} = CPi_{ideal} + MR \times MP \times$ Mem ref / instr

a) miss penalty = 20 × B_{cc} B in block size

B = 8 bytes → 4% MR and MP = 20·8 = 160 CC

$CPi_{real} = 1 + 0,04 \cdot 160 \cdot 1,35 = 1 + 8,64 = 9,64$

B = 16 bytes → 3% MR and MP = 20·16 = 320 CC

$CPi_{real} = 1 + 0,03 \cdot 320 \cdot 1,35 = 13,96$

B = 32 bytes → 2% MR and MP = 20·32 = 640 CC

$CPi_{real} = 1 + 0,02 \cdot 640 \cdot 1,35 = 18,28$

B = 64 bytes → 1,5% MR and MP = 20·64 = 1280 CC

$CPi_{real} = 1 + 0,015 \cdot 1280 \cdot 1,35 = 26,92$

⇒ 8 bytes is the optimal block

b) MP = 24 + B_{cc}

B = 8 bytes → 4% MR and MP = 24+8 = 32 CC

$CPi_{real} = 1 + 0,04 \cdot 32 \cdot 1,35 = 2,728$

B = 16 bytes → 3% MR and MP = 24+16 = 40 CC

$CPi_{real} = 1 + 0,03 \cdot 40 \cdot 1,35 = 2,62$

B = 32 bytes → 2% MR and MP = 24+32 = 56 CC

$CPi_{real} = 1 + 0,02 \cdot 56 \cdot 1,35 = 2,512$

B = 64 bytes → 1,5% MR and MP = 24+64 = 88 CC

$CPi_{real} = 1 + 0,15 \cdot 88 \cdot 1,35 = 2,782$

B = 128 bytes → 1% MR and MP = 24+128 = 152 CC

$CPi_{real} = 1 + 0,01 \cdot 152 \cdot 1,35 = 3,052$

⇒ 32 bytes block is the most efficient

$\boxed{5.10.}$

$MM_{access} = 70$ ns

36% of faster access data memory

	L_1 size	L_1 MR	L_1 Hit time
P_1	2 MiB	8,0%	0,66 ns
P_2	4 MiB	6,0%	0,9 ns

P_1: L_1 MR 8%, L_2 MR 95%, L_2 Hit time = 5,62

AMAT = Hit time + L_1 MR L_2 Hittime + L_1 MR · L_2 MR · MM access time

AMAT = 0,66 + 0,08 · 5,62 + 0,08 · 0,95 · 70 = 6,43 ns

AMAT is worse with L_2 cache

5.10.1

$$CR = \frac{1}{L_1 \text{ Hittime}}$$

P_1 : CR = 1,515 GHz

P_2 : CR = 1,111 GHz

5.10.2 AMAT = HitTime + MR (MP) 70 ns

P_1: AMAT = 6,26 ns

$$AMAT_{CC} = \frac{6,26}{0,66} = 9,48 CC$$

P_2: AMAT = 0,9 + 0,06 · 70 = 5,1 ns

$$AMAT_{CC} = \frac{5,1}{0,9} = 5,67 CC$$

5.10.3

$CPI_{ideal} = 1$ $CPI_{real} = CPI_{ideal} + mem$

P_1: $CPI_{real} = 1 + 0,36(9,48-1) = 4,05$

P_2 : $CPI_{real} = 1 + 0,36(5,67-1) = 2,68$

$\Rightarrow P_2$ is faster

5.10.4

L_2 Size	L_2 MR	L_2 HitTime
1 MiB	95 %	5,62 ns

for P1

$\boxed{5.10.5}$

$CPI_{real} = CPI_{ideal} + \text{Mem access Freq} \cdot (AMAT - 1)$

$AMAT_{CC} = \frac{6,43}{0,66} = 9,74$

$CPI_{real} = 1 + 0,36(9,74-1) = 1 + 0,36 \cdot 8,74 = 4,15$

$\boxed{5.10.6}$

L_2 MR = ? to be faster

$CPI_{L_2} < 4,05$ (old CPI)

$AMAT = 0,66 + 0,08 \cdot 5,62 + 0,08 \cdot MR_{L_2} \cdot 70$

$AMAT_{old} = 6,26$ ns

$\Rightarrow 0,66 + 0,45 + 5,6 \cdot MR_{L_2} < 6,26$

$5,6 MR_{L_2} < 5,15$

$\Rightarrow MR_{L_2} < 0,92$

$\Rightarrow$ the new L_2 MR must be less than 92 %